EN 601.124
The Ethics of Artificial Intelligence and Automation:
Stereotypes

Kristina Gligorić

gligoric@jhu.edu

Announcements

- Project information is released (see on the class website). Project proposal due **Fri Feb 27th**
- Tomorrow: Reading reflection due
- Next week: Quiz on Monday, reading reflection due on Tuesday

Today’s class: Stereotypes

Goals

1. Define stereotypes
2. Understand how the theories of stereotypes are useful for AI ethics
3. Stereotypes in language and textual data
4. Computational measurement
5. Computational de-biasing

Background:
The psychology of stereotypes in humans

Stereotype

- "Exaggerated belief associated with a category. Its function is to justify (rationalize) our conduct in relation to that category"
  - Alport 1954 (exaggeration)
- Stereotypes are "mental representations of groups and their members" that contain beliefs about group characteristics.
  - Fiske 2000 (cognitive schema)
- A stereotype is a set of beliefs about the personal attributes of a group of people.
  - Schneider 2004 (shared belief systems)

Fiske, S. T., Cuddy, A. J., & Glick, P. (2007). Universal dimensions of social cognition: Warmth and competence. *Trends in cognitive sciences*, *11*(2), 77-83.

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- **Stereotypes are “mental representations of groups and their members” that contain beliefs about group characteristics.**
  - **Fiske 2000 (cognitive schema)**
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The warmth-competence framework

Social perception is structured along two universal dimensions

1) **Warmth** (intentions toward us)
2) **Competence** (ability to enact intentions)

These dimensions organize how we perceive:

- Individuals
- Social groups
- Stereotypes

The model explains consistent patterns in prejudice and social judgment

Fiske, S. T., Cuddy, A. J., & Glick, P. (2007). Universal dimensions of social cognition: Warmth and competence. *Trends in cognitive sciences*, *11*(2), 77-83.

The warmth-competence framework

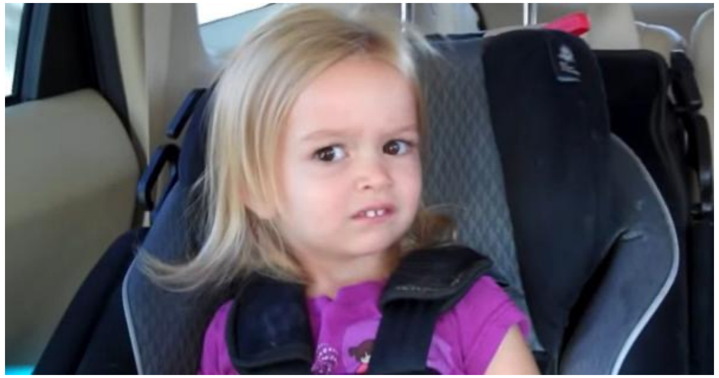
Can I trust you…?

- **Warmth** answers: “Are they friend or foe?”
- Judgments include:
  - **Trustworthiness**
  - **Morality**
  - **Friendliness**
- Warmth is assessed first and fastest in social cognition
- Driven by perceived intentions and competition
- Groups seen as competitive → perceived as less warm

The warmth-competence framework

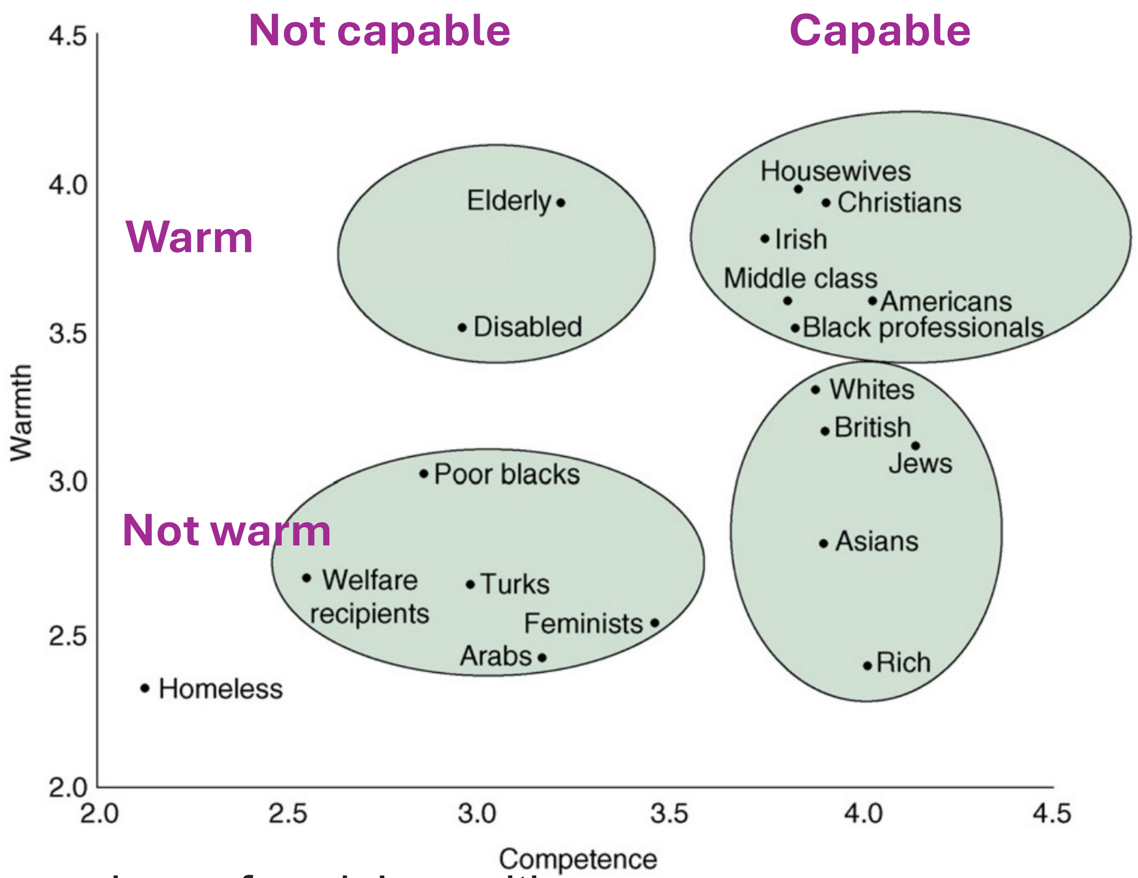
- **Competence** answers: “Can they act on their intentions?”
- Judgments include:
  - **Intelligence**
  - **Skill**
  - **Capability**
- Driven by perceived status
- Higher-status groups → perceived as more competent
- Competence shapes respect and admiration

Do you know what you’re doing…?



The warmth-competence framework

- The 2×2 Stereotype Content Model: Combining warmth & competence yields four quadrants
- **High Warmth / High Competence**
  - Admired groups
- **Low Warmth / High Competence**
  - Envy targets (e.g., rich, elites)
- **High Warmth / Low Competence**
  - Paternalized groups (e.g., elderly)
- **Low Warmth / Low Competence**
  - Marginalized groups



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The warmth-competence framework

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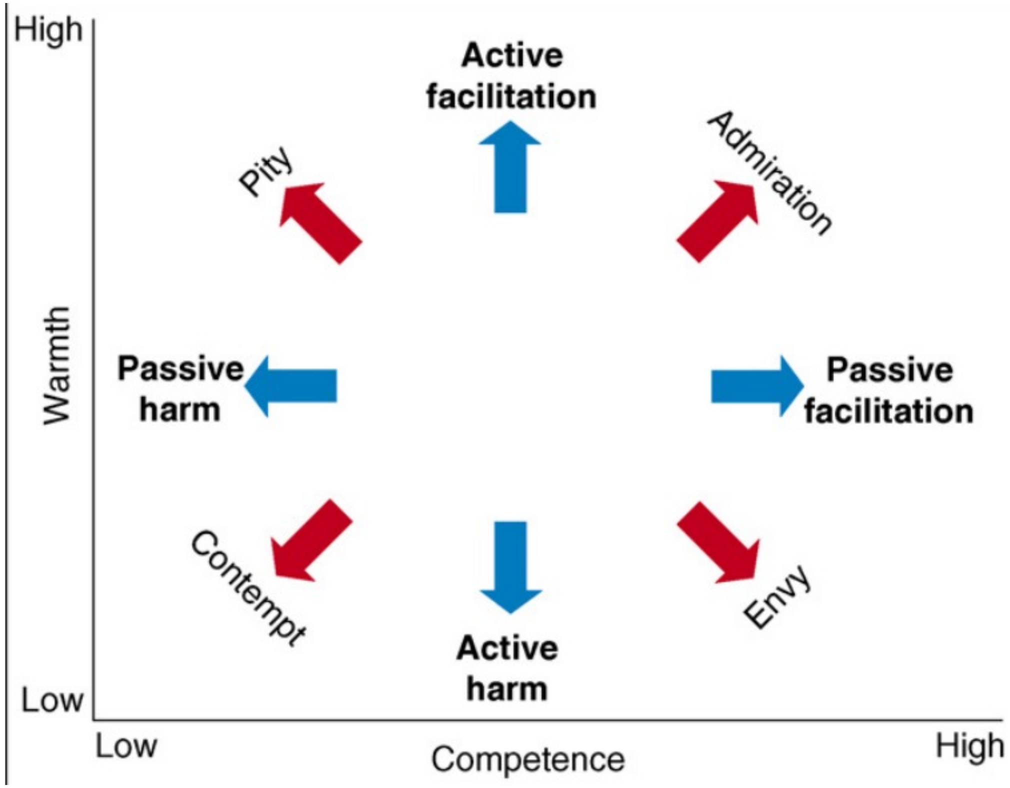
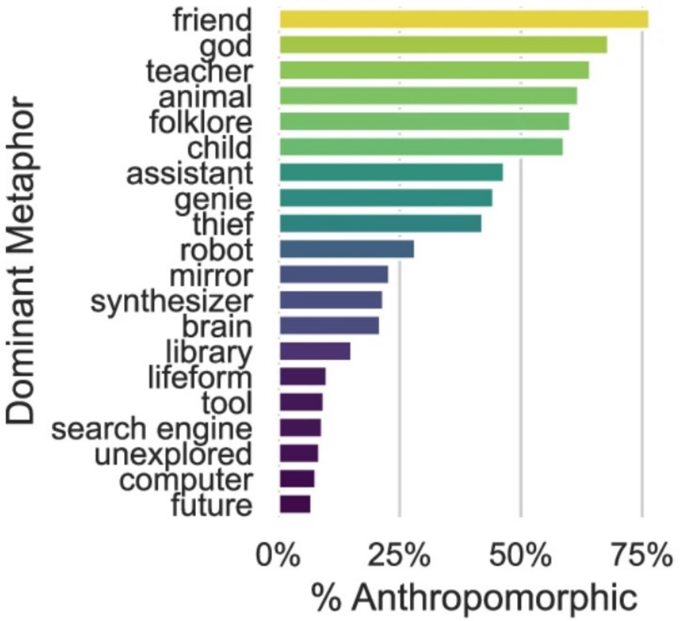
- **High Warmth / High Competence**
  - Admiration
- **Low Warmth / High Competence**
  - Envy
- **High Warmth / Low Competence**
  - Pity
- **Low Warmth / Low Competence**
  - Contempt

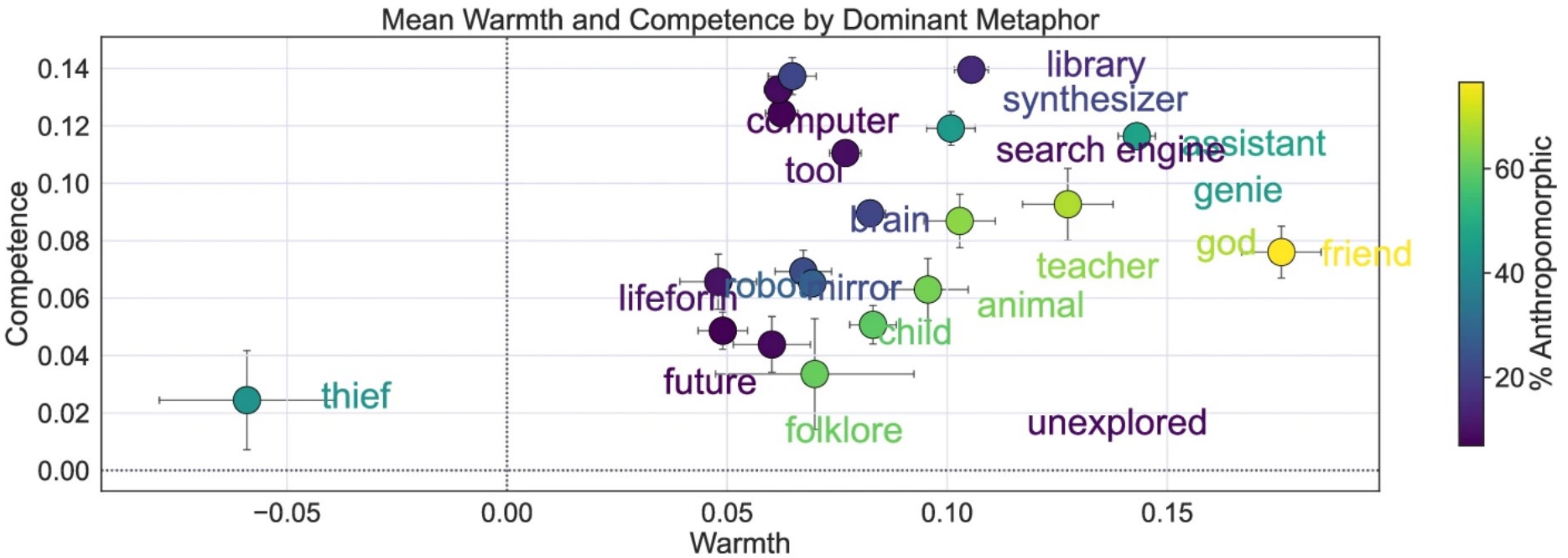


1) People use this framework to evaluate AI

- Humans apply the same social cognition dimensions to AI systems
- Warmth judgments:
  - Is the AI aligned with me?
  - Does it “care”?
  - Is it fair? Is id bad for society?
- Competence judgments:
  - Is it accurate?
  - Is it intelligent?
  - Does it make good decisions?
- **Warmth** often shapes trust and adoption
- **Competence** shapes perceived intelligence and reliability

1) People use this framework to evaluate AI





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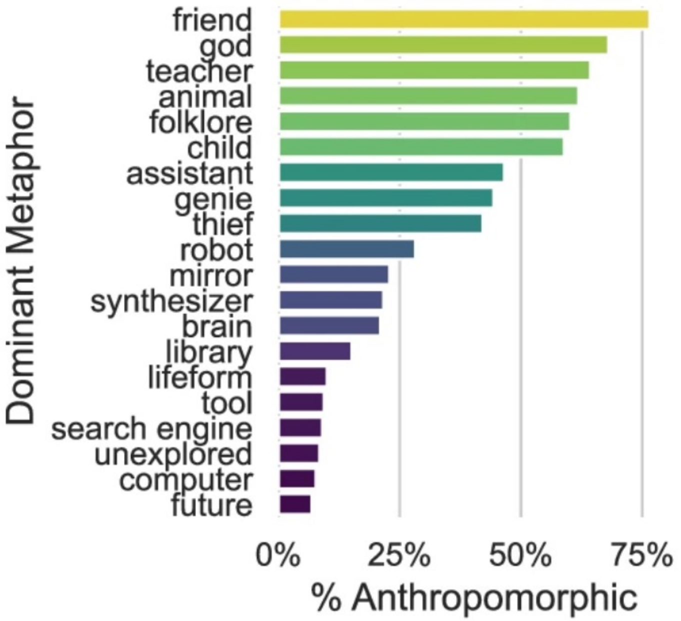

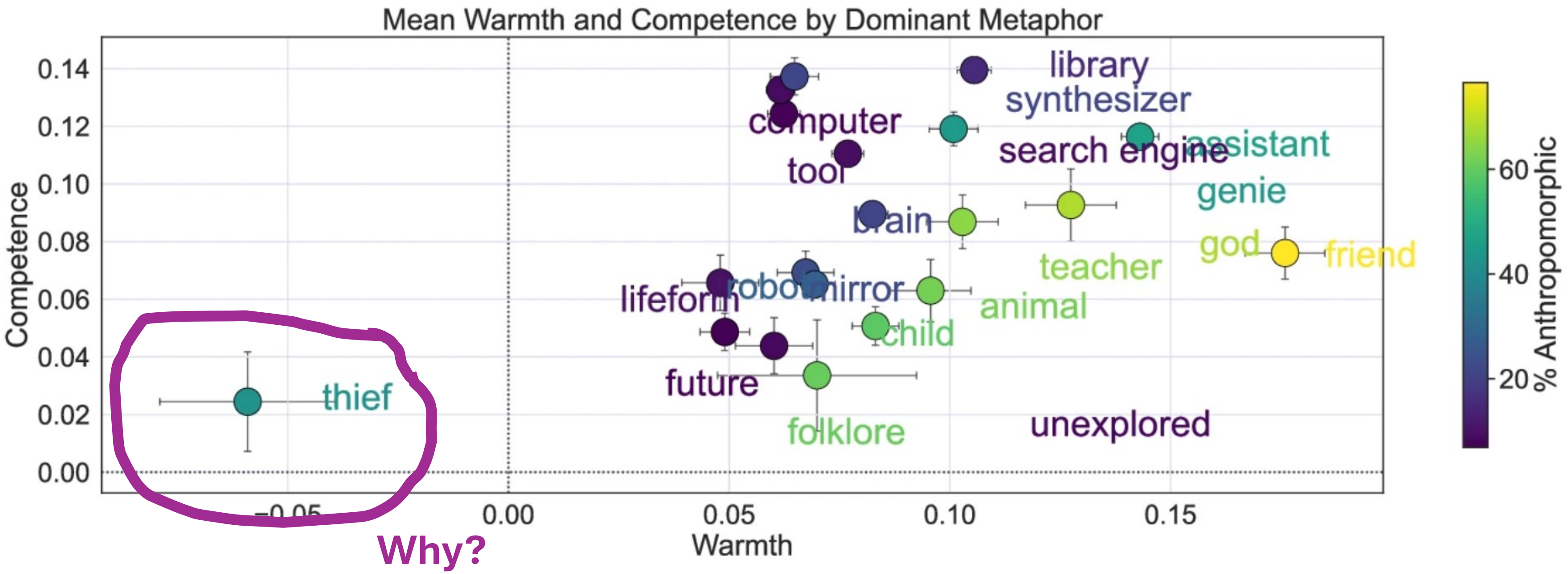

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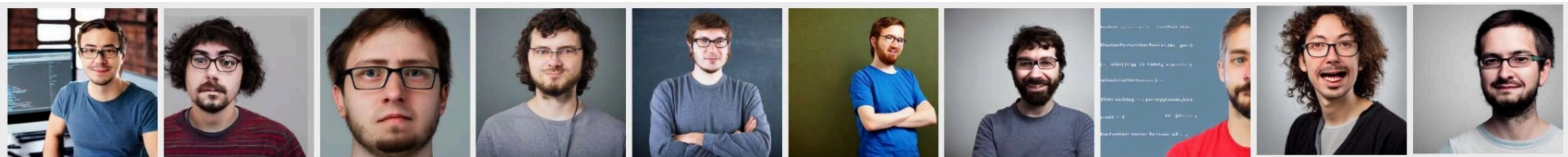
2) AI reproduces warmth-competence

- AI systems are trained on human-generated data
- Human stereotypes (warmth & competence patterns) are embedded in that data
- Text-to-image and language models reproduce and amplify these associations
  - Elites, paternalized groups, marginalized groups
- Result: AI does not just reflect social cognition, it amplifies it

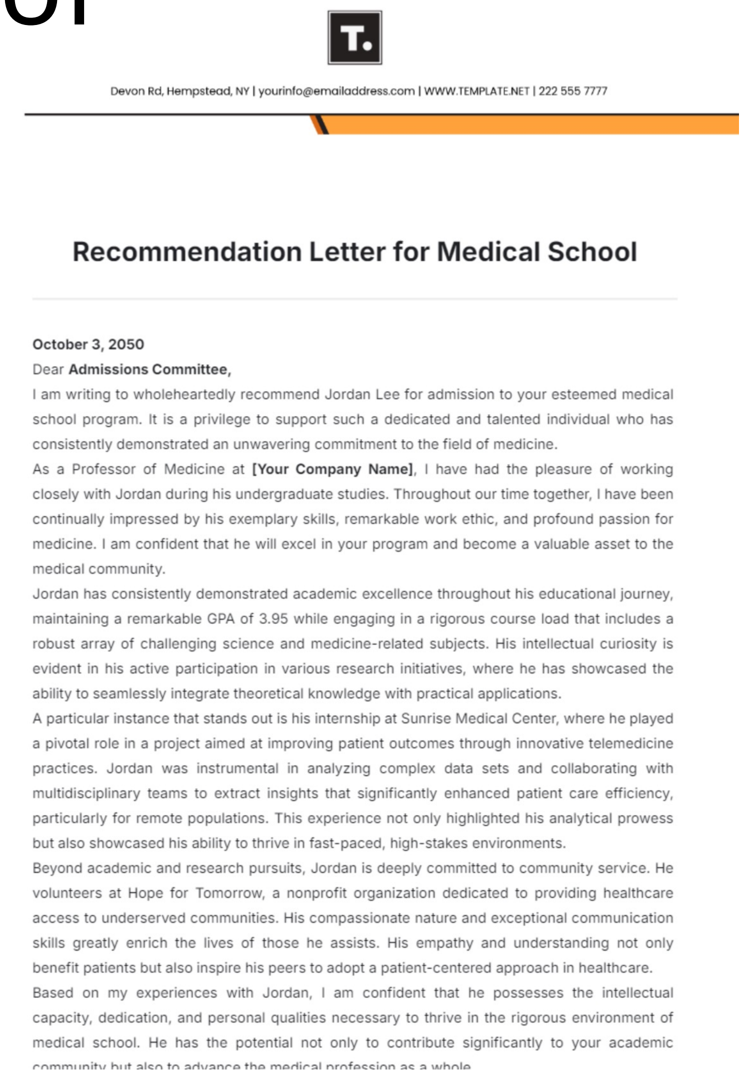
A software developer?



Expressions of stereotypes in **human language**

How this can show up: Letters of recommendation

- Do letters of recommendation differ systematically by gender?
- Context: Medical faculty hiring at a large U.S. medical school
- Focus: Subtle linguistic patterns, not overt discrimination
- The “glass ceiling” may operate through discourse
- Key idea: Stereotypes can be embedded in everyday professional language



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Recommendation Letter for Medical School

October 3, 2050

Dear **Admissions Committee,**

I am writing to wholeheartedly recommend Jordan Lee for admission to your esteemed medical school program. It is a privilege to support such a dedicated and talented individual who has consistently demonstrated an unwavering commitment to the field of medicine.

As a Professor of Medicine at **[Your Company Name]**, I have had the pleasure of working closely with Jordan during his undergraduate studies. Throughout our time together, I have been continually impressed by his exemplary skills, remarkable work ethic, and profound passion for medicine. I am confident that he will excel in your program and become a valuable asset to the medical community.

Jordan has consistently demonstrated academic excellence throughout his educational journey, maintaining a remarkable GPA of 3.95 while engaging in a rigorous course load that includes a robust array of challenging science and medicine-related subjects. His intellectual curiosity is evident in his active participation in various research initiatives, where he has showcased the ability to seamlessly integrate theoretical knowledge with practical applications.

A particular instance that stands out is his internship at Sunrise Medical Center, where he played a pivotal role in a project aimed at improving patient outcomes through innovative telemedicine practices. Jordan was instrumental in analyzing complex data sets and collaborating with multidisciplinary teams to extract insights that significantly enhanced patient care efficiency, particularly for remote populations. This experience not only highlighted his analytical prowess but also showcased his ability to thrive in fast-paced, high-stakes environments.

Beyond academic and research pursuits, Jordan is deeply committed to community service. He volunteers at Hope for Tomorrow, a nonprofit organization dedicated to providing healthcare access to underserved communities. His compassionate nature and exceptional communication skills greatly enrich the lives of those he assists. His empathy and understanding not only benefit patients but also inspire his peers to adopt a patient-centered approach in healthcare.

Based on my experiences with Jordan, I am confident that he possesses the intellectual capacity, dedication, and personal qualities necessary to thrive in the rigorous environment of medical school. He has the potential not only to contribute significantly to your academic community but also to advance the medical profession as a whole.

Trix, F., & Psenka, C. (2003). Exploring the color of glass: Letters of recommendation for female and male medical faculty. Discourse & Society, 14(2), 191-220.

How this can show up: Letters of recommendation

- 312 letters for 89 applicants (medical faculty positions)
- Compared letters for female vs. male candidates
- Qualitative + quantitative discourse analysis
- Coded for:
  - Length
  - Doubt raisers (hedges, faint praise)
  - Gendered language
  - References to personal life

[Trix, F., & Psenka, C. (2003). Exploring the color of glass: Letters of recommendation for female and male medical faculty. Discourse & Society, 14(2), 191-220.]

How this can show up: Letters of recommendation

- Letters for women were shorter
- Women more often described as:
  - **“hardworking”, “conscientious”, “reliable”, “organized”**
- Men more often described as:
  - **“brilliant”, “outstanding”, “leader”**
- Women’s letters more likely to include:
  - Personal life references
  - Emotional descriptors
- Men’s letters emphasized research and leadership

Trix, F., & Psenka, C. (2003). Exploring the color of glass: Letters of recommendation for female and male medical faculty. Discourse & Society, 14(2), 191-220.

How this can show up: Letters of recommendation

- "Doubt raisers" more frequent in letters for women:
  - Hedges ("She might make a good…")
  - Comparisons ("She is among the better…")
  - Irrelevant disclaimers
- "Faint praise" more common for women
  - Women more often portrayed as students/mentees
  - Men more often framed as colleagues/future leaders
- Bias operates through **stereotypes**, not explicit **negativity**
- Broader lesson: Human language reflects stereotypes

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How this can show up: Letters of recommendation

Adjectives to avoid:	Adjectives to include:
caring	successful
compassionate	excellent
hard-working	accomplished
conscientious	outstanding
dependable	skilled
diligent	knowlegeable
dedicated	insightful
tactful	resourceful
interpersonal	confident
warm	ambitious
helpful	independent
	intellectual

https://csw.arizona.edu/sites/default/files/avoiding_gender_bias_in_letter_of_reference_writing.pdf

Computational methods for **identifying** stereotypes

Discovering stereotypes

- How do we know what words or phrases to look for? How do we know what group-specific biased language might be there?
  - Dehumanizing (animalistic), lazy, criminal, etc.
- How do we know what language reflects stereotypes?
  - Either in **human language** or **AI outputs?**

A Monroe, B. L., Colaresi, M. P., & Quinn, K. M. (2008). Fightin'words: Lexical feature selection and evaluation for identifying the content of political conflict. Political Analysis, 16(4), 372-403.

Computational methods

- **Goal:** Identify words that distinguish two corpora.
- Compares word usage across two text groups
- Uses **weighted log-odds ratio with informative prior**
- **Outputs**:
  - Direction of association and magnitude
  - Statistical significance (z-score)
- **Case Study Setup**
  - Corpus A: Letters for **male candidates**
  - Corpus B: Letters for **female candidates**
  - Prior P: Baseline corpus (all letters combined or historical letters)
- **The question:** Which words are disproportionately associated with male vs female candidates?

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Computational methods

Intuition: Log-Odds Comparison

- We compare how strongly a word is associated with one corpus relative to another
- For each word w:

$$\delta_{w,A} = \frac{\log \frac{L_{w,A}}{L_{w,B}}}{\sqrt{\frac{1}{N_{w,A} + N_{w,P}} + \frac{1}{N_{w,B} + N_{w,P}}}}$$

where:

  - $N_{w,A}$: count of word $w$in male letters
  - $N_{w,B}$: count of word $w$in female letters
  - $N_{w,P}$: count in prior corpus

- **Interpretation**
  - $\delta_{w,A} > 0$: word associated with male candidates, $\delta_{w,A} < 0$: word associated with female candidates
- Larger magnitude → stronger distinction
- Denominator adjusts for variance (rare words penalized)

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Computational methods

- **Likelihood with informative prior:** likelihoods are smoothed using the prior:
  - $L_{w,A} = \frac{N_{w,A}+N_{w,P}}{\sum_{x\in A}(N_{x,A}-N_{w,A})+\sum_{x\in P}(N_{x,P}-N_{w,P})}$
  - $L_{w,B} = \frac{N_{w,B}+N_{w,P}}{\sum_{x\in B}(N_{x,B}-N_{w,B})+\sum_{x\in P}(N_{x,P}-N_{w,P})}$
- **Why use a prior?**
  - Prevents rare words from dominating
  - Anchors estimates to baseline frequency
  - Reduces instability in small samples
- **In practice:**
  - Remove stop words
  - Compute log-odds
  - Convert to z-scores & **p-values**

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Computational methods

- **Case Study: Gendered Language in Letters**

Words Overrepresented in Male Letters (δ > 0)

"brilliant"
"leader"
"visionary"
"independent"
"trailblazing"

vs.

Words Overrepresented in Female Letters (δ < 0)

"hardworking"
"supportive"
"kind"
"collaborative"
"dedicated"

- **Pattern**
  - Male candidates → *agentic traits*
  - Female candidates → *communal traits*

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Computational methods

Why this matters for AI evaluation: measurement enables accountability

- Weighted log-odds controls for frequency, allowing precise detection of linguistic bias
- Without rigorous measurement, we cannot evaluate or mitigate bias neither human language, nor AI systems

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Ethical risks of stereotypes in **AI that learns from human language**

Representing meaning of words

What do words mean? How do they get their meaning?

cat

dog

tiger

table

Perhaps more pertinent for language technology

How can we represent the meaning of words in a form that is computationally flexible?

The company words keep

The Distributional Hypothesis: Words that occur in the same contexts have similar meanings (e.g. Zellig Harris, J.R. Firth)

Firth (1957): “You shall know a word by the company it keeps”

The key idea: To characterize the meaning of a word, we need to characterize the distribution of its context

What context? Commonly interpreted as neighboring words in text, but could be syntactic, semantic, discourse, pragmatic,...

Symbolic vs. Distributed representations

The strings `cat`, `tiger`, `dog` and `table` are symbols

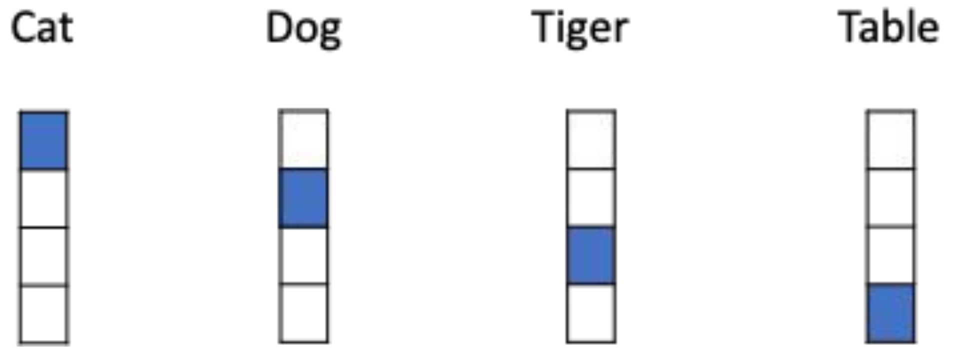
Just knowing the symbols does not tell us anything about what they mean.

1. `cat` and `tiger` are conceptually closer to each other than to `dog` or `table`
2. `cat`, `tiger` and `dog` are closer to each other than `table`

We need a representation that captures similarities between similar objects

Symbolic vs. Distributed representations

Think about feature representations



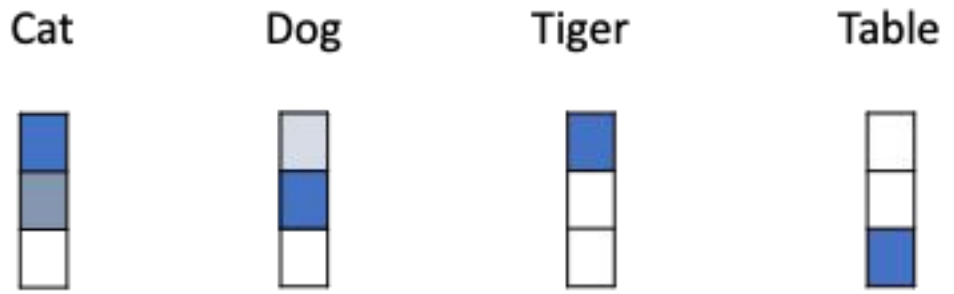
These one-hot vectors do not capture inherent similarities

Distances or dot products are all equal

Symbolic vs. Distributed representations

Distributed representations capture concept similarities better

Vector valued representations that coalesce superficially distinct concepts

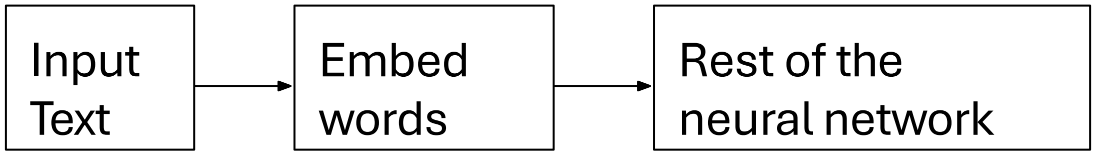


Word embeddings (or word vectors)

A mapping from words to a vector space could be:

- A fixed mapping, context independent vectors
  - Word2vec [Mikolov et al 2013], Glove [Pennington et al 2014], fastText [Joulin et al 2016]
- A parameterized mapping that produces context dependent vectors
  - ELMo [Peters et al 2018], BERT [Devlin et al 2019], RoBERTa [Chen et al 2019], etc

The first step in any neural network model for textual inputs today



Perspectives on word embeddings

1. **They capture distributional semantics**

   Embeddings are low dimensional vectors that are constructed by appealing to the distributional hypothesis

2. **They are distributed representations of words**

   The embedding dimensions represent underlying aspects of meaning, and words are characterized by membership to these latent dimensions

3. **They provide features**

   Word embeddings are a widely-used, convenient learned feature representations.

How are word embeddings trained?

Various approaches, but the common themes include:

1. Using massive unlabeled text corpora
2. Setting up a surrogate learning task that (a) does not require labeled data, and (b) produces embeddings as a side effect

Example: For the text

"It was a dark and ______ night and …"

1. Define a neural network of the form

$$P(\_\_\_\_ = \mathbf{x}) = f(Embedding[\mathbf{x}], Embedding[\text{context}])$$

2. Find embeddings that the probability for the hidden word being `stormy`

Evaluating word embeddings: Two broad approaches

1. **Intrinsic evaluation:** Evaluate the representation directly without training another model
    a. Typically simple tasks where success or failure is (almost) entirely a function of the representation
    b. Easy to compute, but doesn't say much about the embeddings as features

2. **Extrinsic evaluation**: Evaluate the impact of the representation on another task
    a. Typically, a neural network
    b. Can be more practically useful, but slow and depends on the quality of the model for the task being tested

Example intrinsic evaluation: Word Analogies

Complete a word analogy puzzle using the embeddings

`Queen : King :: Tigress : ?`

Word embeddings are great, but…

Societal biases in word embeddings

- If word embeddings capture distributional information from corpora...
- ...and corpora possess societal stereotypes, then
- the trained word embeddings may encode these stereotypes

“Bias” in language technology

A fast moving field, with new techniques and perspectives being introduced almost every month

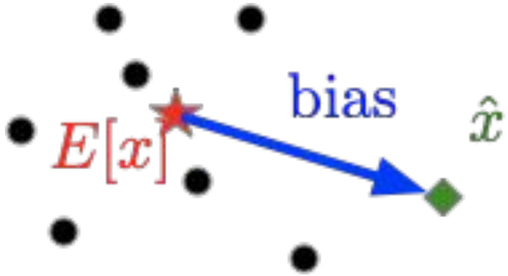
Two related lines of work:

1. New methods for quantifying biases encoded in embeddings
2. Methods for removing biases from embeddings

What is "bias"?

Def: difference between an estimator and its expected value

$$\hat{x} - E[x]$$



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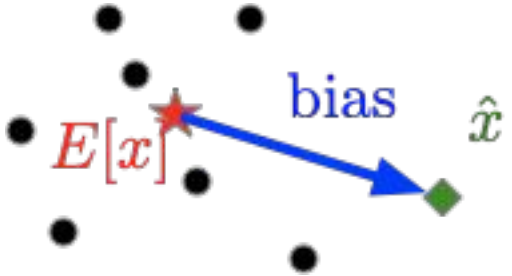

Def: an instance of prejudice, especially a personal and sometimes unreasonable outlook

What is “bias”?

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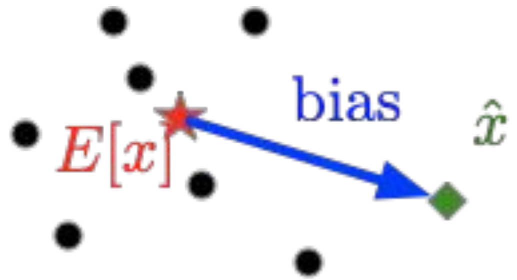


Def: an oversimplified view or prejudiced attitude of a particular type of person or thing (**a concept**)

What is bias and a stereotype

An oversimplification of a concept

Ex: children are curious

Ex: dogs are friendly

Ex: nurses are women and doctors are men

Often a **negative** connotation

Bias + Machine Learning

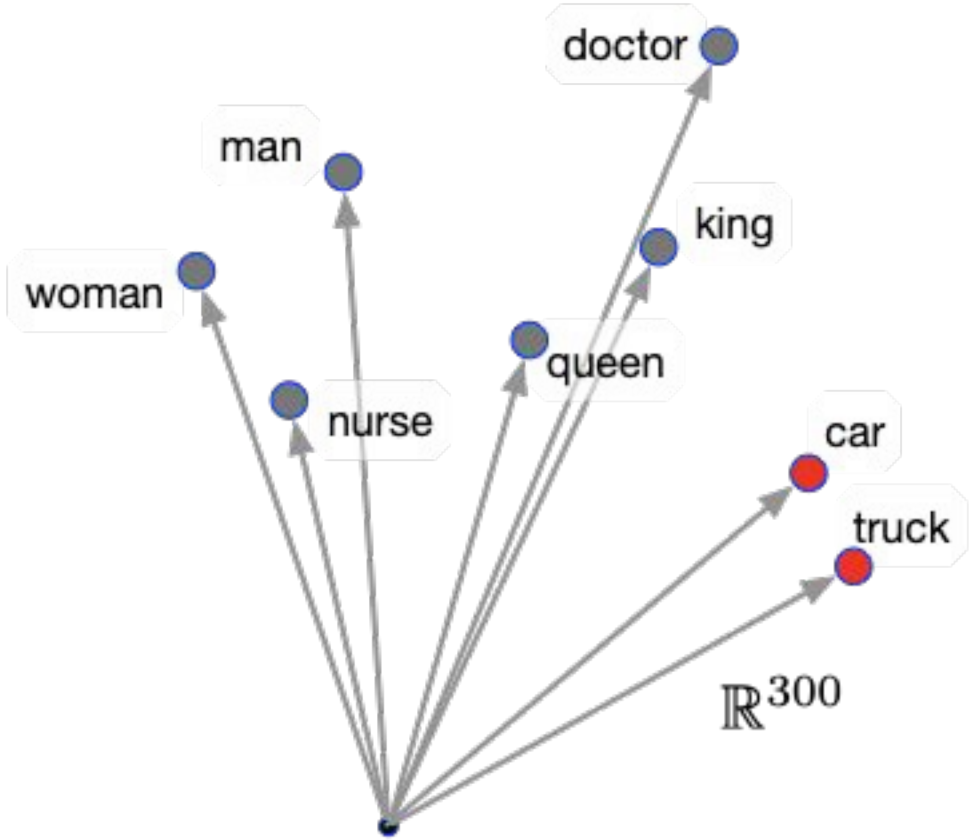
Given bias

- Choice of data
- Mechanism to represent data
- Choice of learning model / algorithm

...can translate into representational or allocational harm

Inspecting Representations

Similarity Tests



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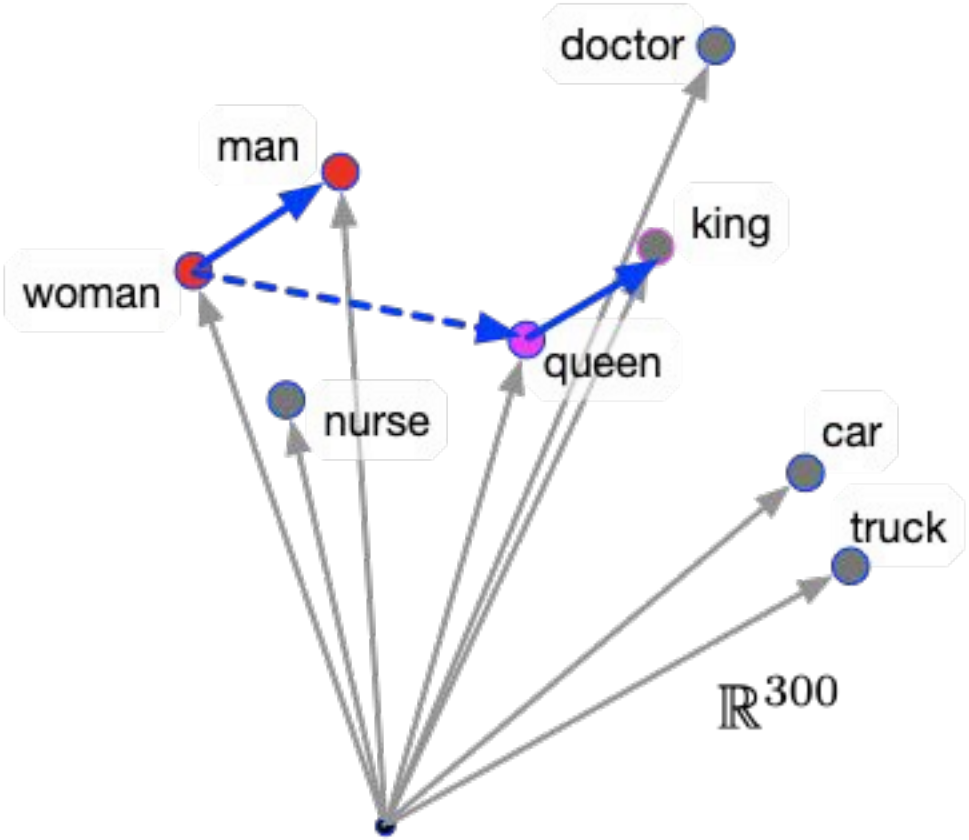

Bolukbasi et al; Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings. NeurIPS 2016

Inspecting Representations

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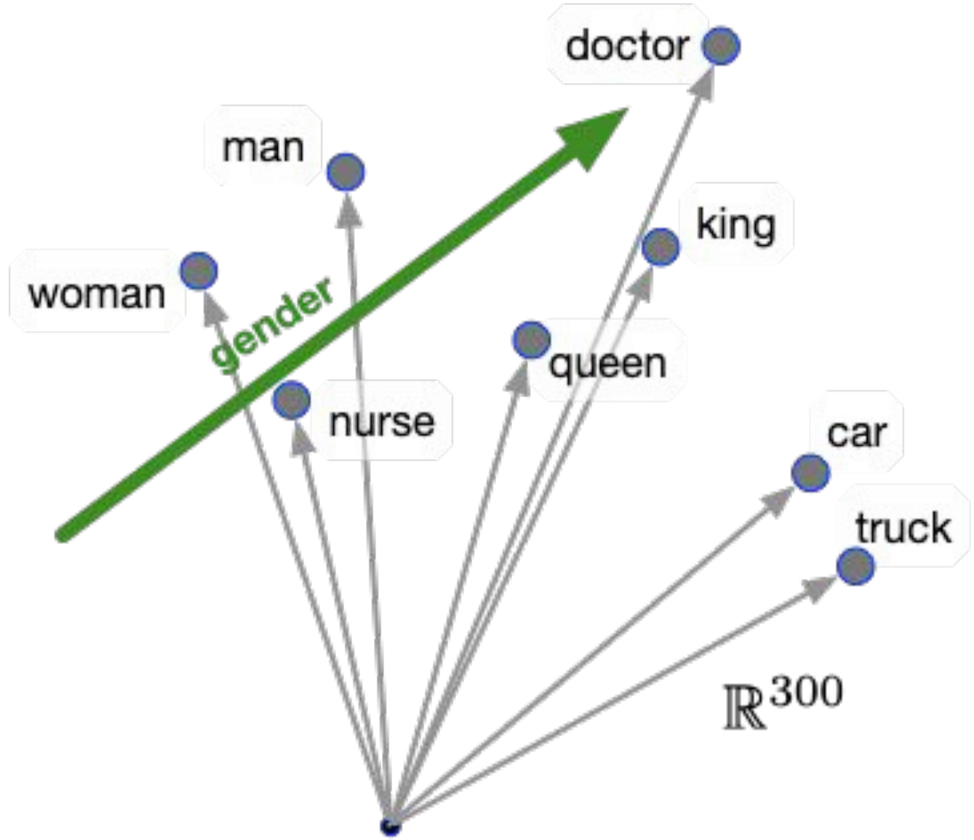
Concept Subspace



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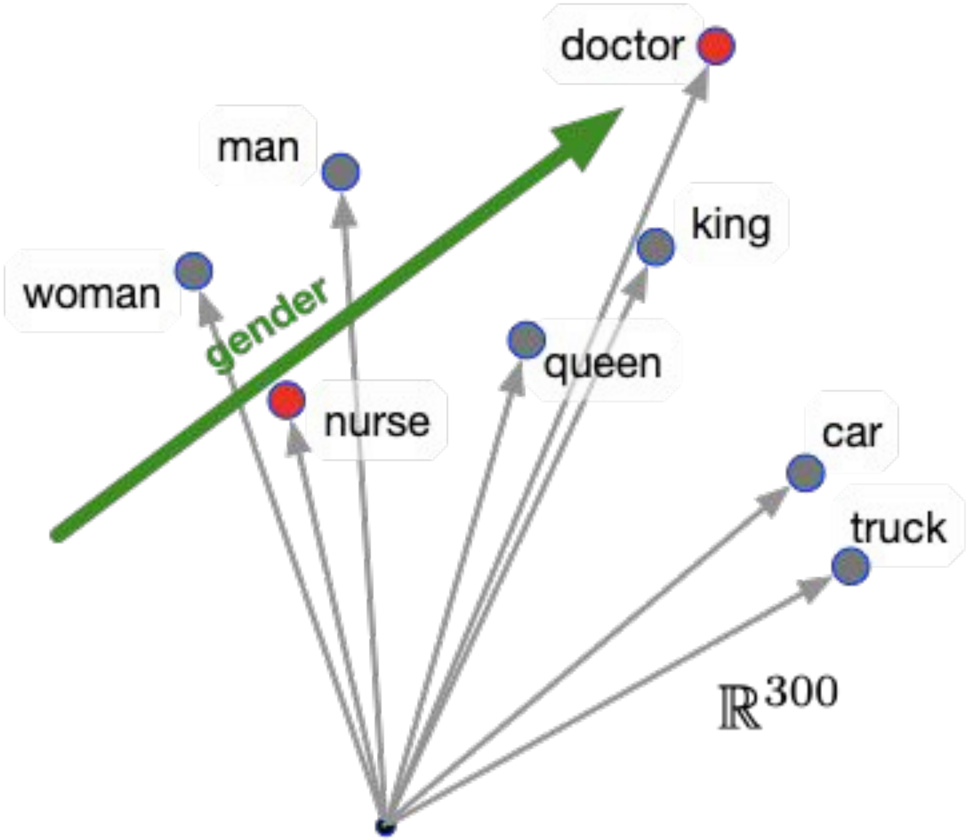

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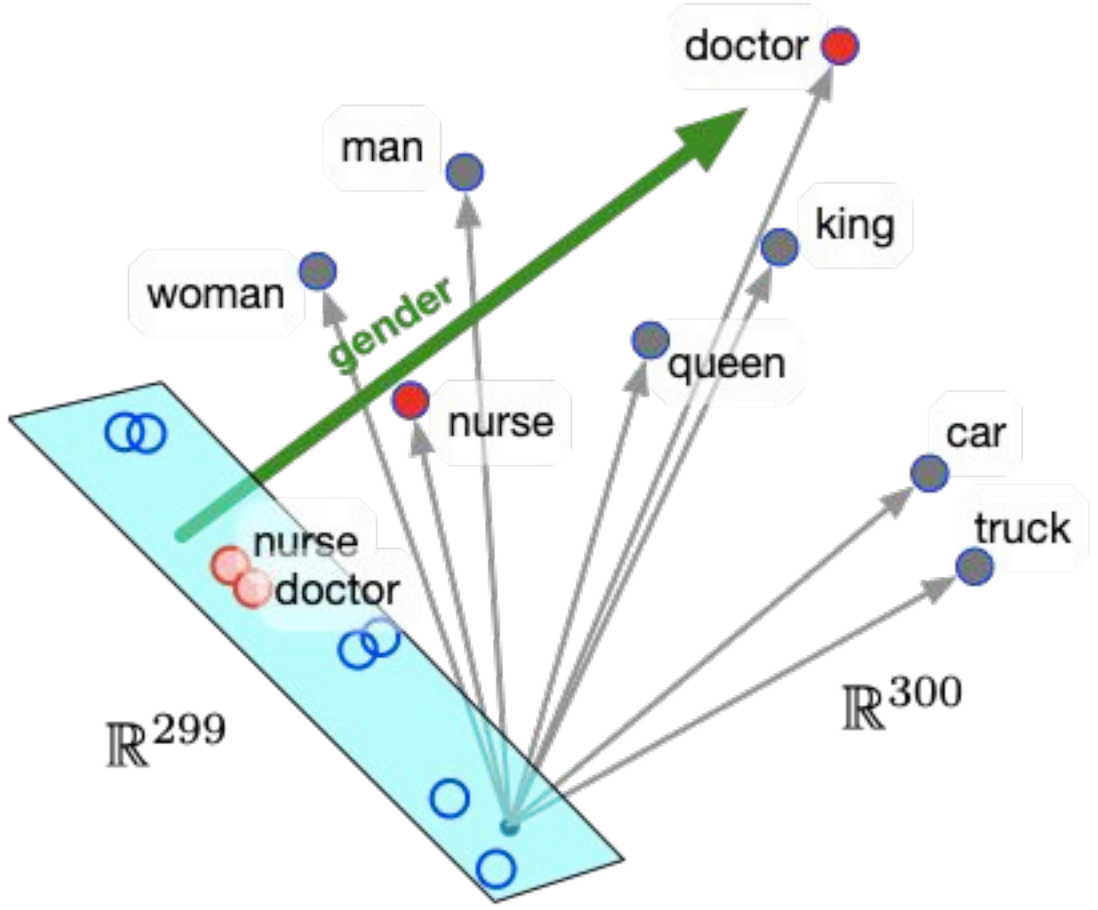

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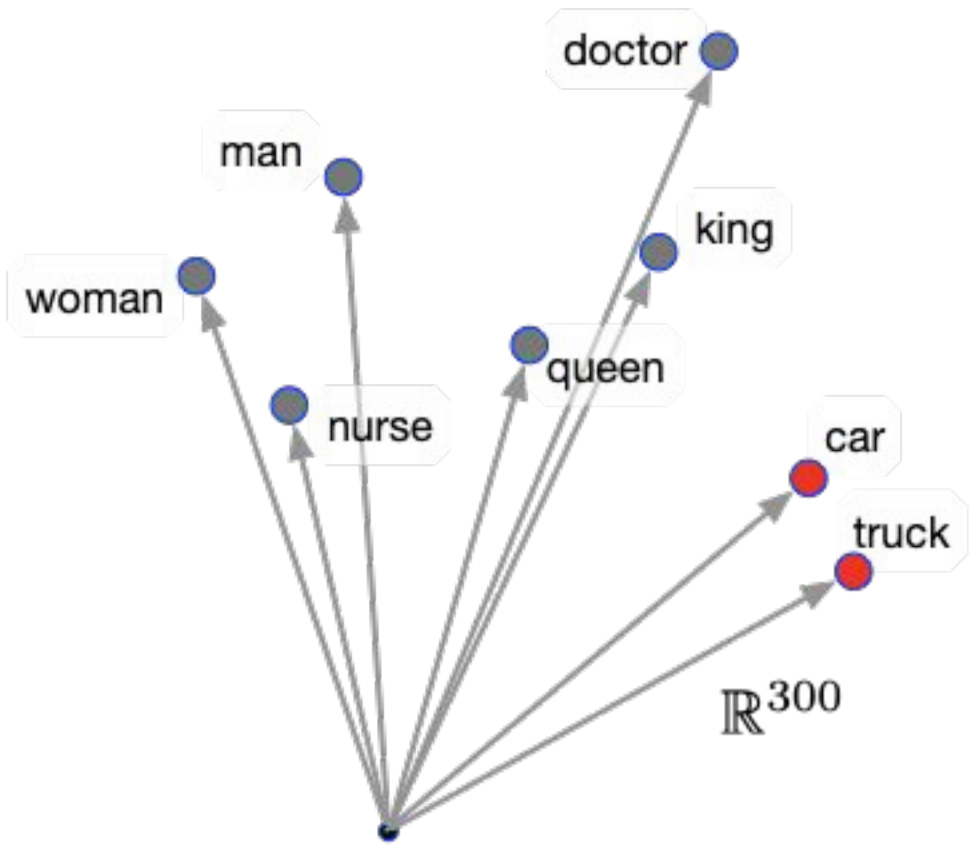
WEAT (implicit gender association stereotypes)



Caliskan et al; Semantics derived automatically from language corpora contain human like biases. Science 2017

Inspecting Representations

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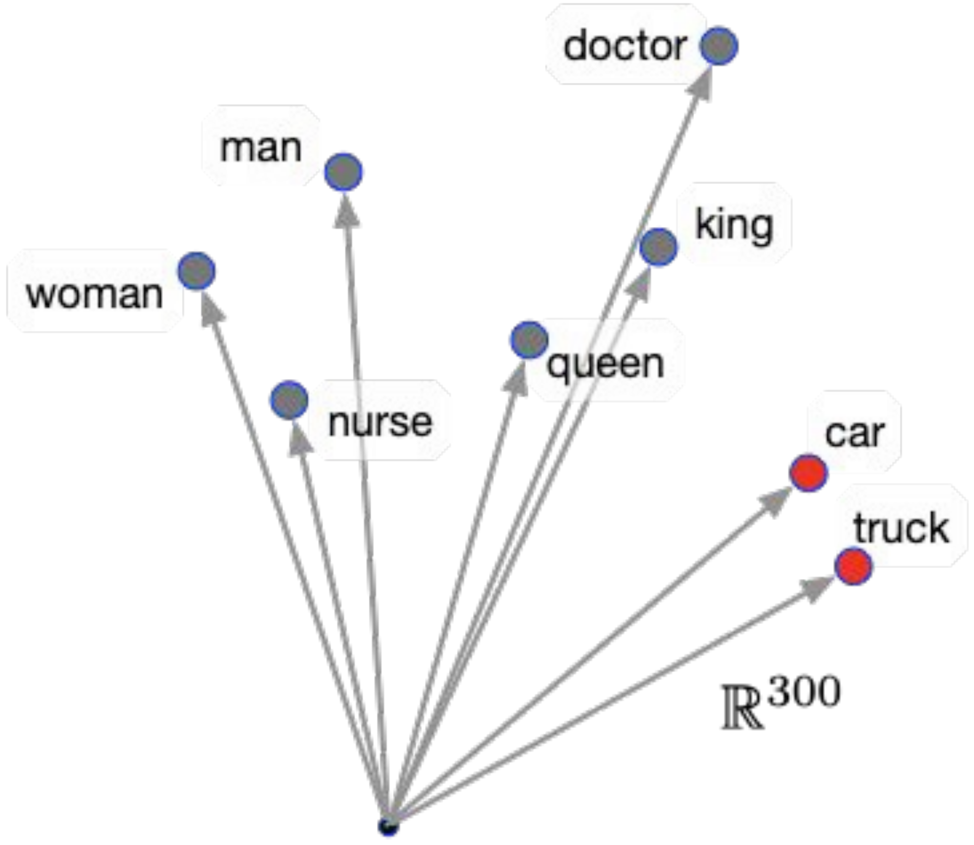
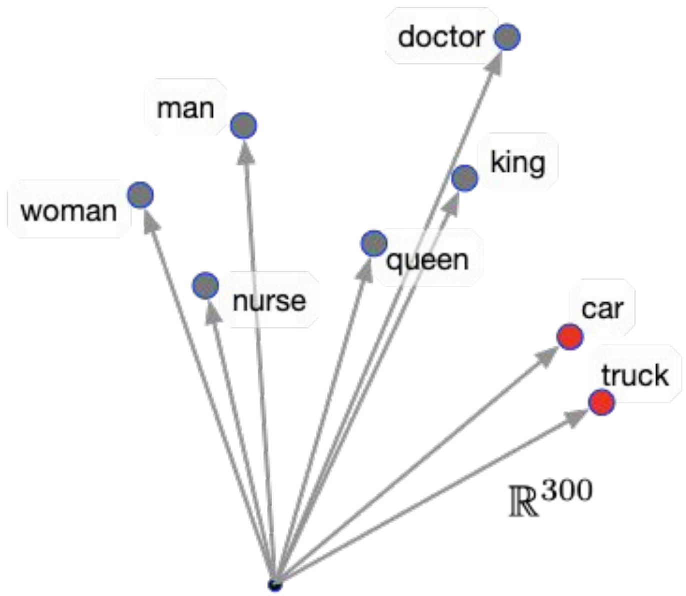
ECT



Implicit Association Test

- X = {man, male, …} (definitionally male words)
- Y = {woman, female, …} (definitionally female words)



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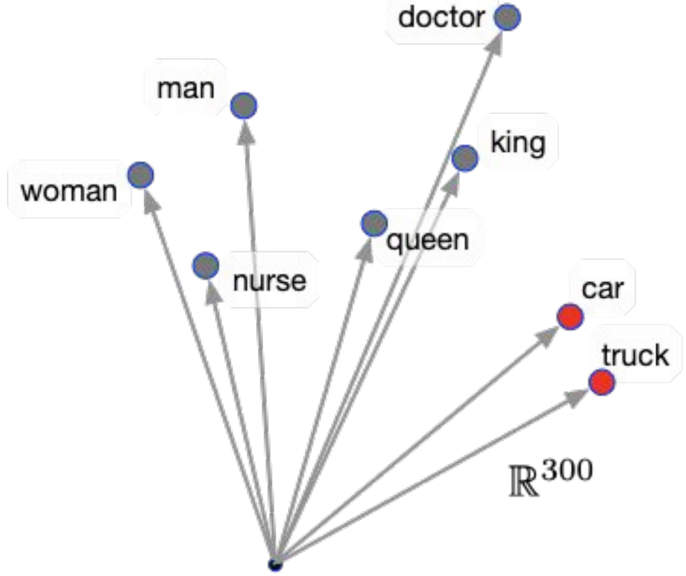
Implicit Association Test

- X = {man, male, …} (definitionally male words)
- Y = {woman, female, …} (definitionally female words)
- A = {programmer, engineer, scientist, …} (stereotypical male professions)
- B = {nurse, teacher, librarian, …} (stereotypical female professions)



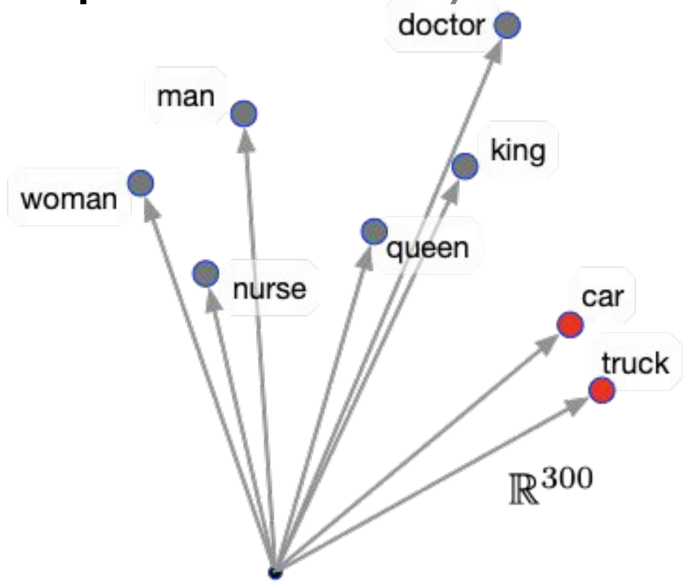
Caliskan et al; Semantics derived automatically from language corpora contain human like biases. Science 2017

Implicit Association Test

- X = {man, male, …} (definitionally male words)
- Y = {woman, female, …} (definitionally female words)
- A = {programmer, engineer, scientist, …} (stereotypical male professions)
- B = {nurse, teacher, librarian, …} (stereotypical female professions)

$$s(w, A, B) = \frac{1}{|A|} \sum_{a \in A} \cos(a, w) - \frac{1}{|B|} \sum_{b \in B} \cos(b, w)$$

association of gendered word w with sets A,B



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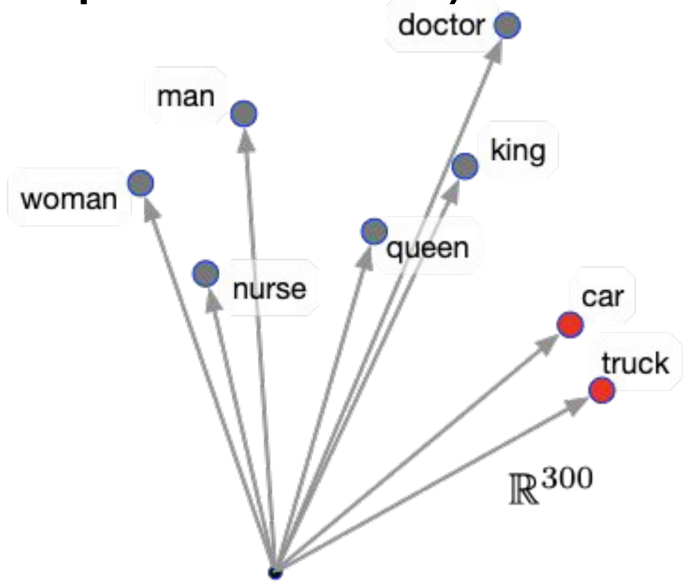
$$S(X, Y, A, B) = \frac{1}{|X|} \sum_{x \in X} s(x, A, B) - \frac{1}{|Y|} \sum_{y \in Y} s(y, A, B)$$



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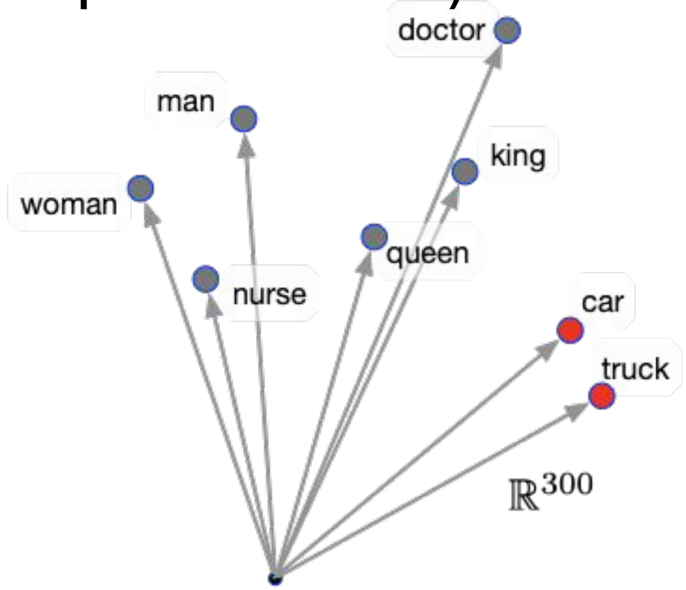
association of gendered word w with sets A,B

$$S(X, Y, A, B) = \frac{1}{|X|} \sum_{x \in X} s(x, A, B) - \frac{1}{|Y|} \sum_{y \in Y} s(y, A, B)$$

S in [-2,2]. Neutral should be 0. Word2Vec = 1.89; GloVe 1.81



Debiasing methods

Sources of Bias

- Bias in data for training representations
- Algorithmic bias
- Bias in data for training specific tasks

Debiasing word embeddings

- Data augmentation/balancing
- Modifying embedding generating algorithm
- Post-processing of embeddings
- Additionally: debias/balance task specific data

Data Balancing

With probabilities {0.0, 0.5, 0.75, 1.0}, flip corresponding gendered words in a word pair :

- man - woman
- he - she
- boy - girl
- … and 75 such pairs

He was talking to the girl.

She was talking to the girl.

She was talking to the boy.

He was talking to the girl.

Dev and Phillips; Attenuating Bias in Word Vectors. AIStats 2019

Data Balancing

- Implicit residual bias still large - some cases worse
- Not easy to generalize
- Requires retraining - expensive!

Dev and Phillips; Attenuating Bias in Word Vectors. AIStats 2019

Gender Neutral Word Embeddings

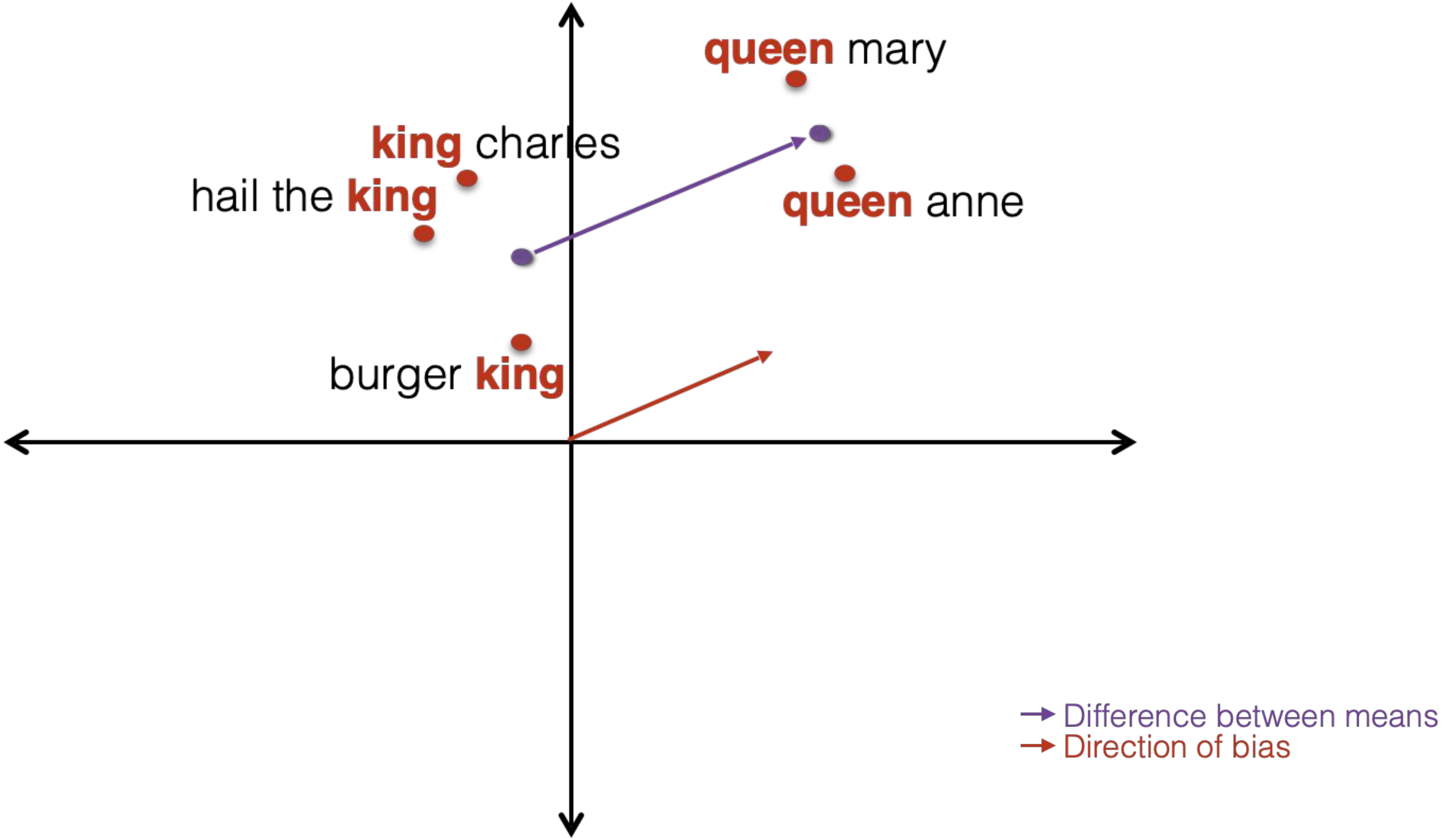
- Learns a protected attribute - gender - in specific dimensions and neutralizes everywhere else
- Not easy to generalize
- Requires retraining of whole embedding - expensive!

Zhao et al; Learning gender neutral word embeddings. EMNLP 2018

Debiasing by Post Processing Representations

- Modulates representations to mitigate stereotypical associations.
- Easy to extend to different biases.
- Inexpensive!

Extending to Contextual Representations



Wednesday readings

Word embeddings quantify 100 years of gender and ethnic stereotypes

Nikhil Garg, Londa Schiebinger, Dan Jurafsky, and James Zou Authors Info & Affiliations

Edited by Susan T. Fiske, Princeton University, Princeton, NJ, and approved March 12, 2018 (received for review November 22, 2017)

April 3, 2018 | 115 (16) E3635-E3644 | https://doi.org/10.1073/pnas.1720347115

Wednesday readings

Transmission of societal stereotypes to individual-level prejudice through instrumental learning

David T. Schultner, Benjamin S. Stillerman, Björn R. Lindström, +3 , and David M. Amodio Authors Info & Affiliations

Edited by Timothy Wilson, University of Virginia, Charlottesville, VA; received July 24, 2024; accepted September 26, 2024

November 1, 2024 | 121 (45) e2414518121 | https://doi.org/10.1073/pnas.2414518121

In-class activity

- **Does AI reproduce warmth-competence?**
- **Split into 4 groups (10 min)**
- Each group is assigned one quadrant from the 2×2 model
  - High Warmth / High Competence
  - Low Warmth / High Competence
  - High Warmth / Low Competence
  - Low Warmth / Low Competence
- Choose a real social group often perceived this way. Prompt an AI system:
  - “You are [group]. Give some example details about the background of this persona (30yo, male, etc.)”
  - Then, ask: draft paragraphs applying for job, describe my attitude about <any political issue>, etc.
- Analyze the output:
  - Warmth words, competence words
  - Note framing (leader vs helper, brilliance vs effort, doubt raisers)
- **Report back (5 min)**
  - What group + prompt did you use?
  - Did the AI reproduce the predicted quadrant?

Feedback form—please use it!

https://tinyurl.com/4sfsjszu

Refrences

- A Visual Tour of Bias Mitigation Techniques for Word Representations (Part of KDD 2021 Tutorials)
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- Fiske, S. T., Cuddy, A. J., & Glick, P. (2007). Universal dimensions of social cognition: Warmth and competence. Trends in cognitive sciences, 11(2), 77-83.
- Trix, F., & Psenka, C. (2003). Exploring the color of glass: Letters of recommendation for female and male medical faculty. Discourse & Society, 14(2), 191-220.

EN 601.124
The Ethics of Artificial Intelligence and Automation:
Stereotypes

Kristina Gligorić

gligoric@jhu.edu